

# Retail Demand Forecasting

## Overall Process Flow Document

### 1. Data Ingestion

I loaded the **train**, **test**, **features**, and **stores** datasets using the Pandas library. I verified the dataset structure by checking dimensions, column names, data types. Identified the datasets for missing values and duplicate records. Also merged the datasets using **Store**, **Date**, and **IsHoliday** as common keys. Prepared a unified dataset for subsequent preprocessing and analysis.

### 2. Data Cleaning & Transformation

I handled missing values using appropriate imputation techniques. Converted the **Date** column into datetime format for time-based feature extraction. Also Encoded categorical variables such as **Store Type** using one-hot encoding. Converted the **IsHoliday** column into a numerical format. I Applied feature scaling where required for Linear Regression.

### 3. Feature Engineering Pipeline

Extracted Multiple columns from date feature, they are **Year**, **Month**, and **Week**. And I Created **Total\_MarkDown** by combining all markdown variables. Calculated aggregated features named **Store\_Avg\_Sales** and **Dept\_Avg\_Sales**. Developed interaction features such as **Temp\_Fuel** and **Economic\_Index**. Prepared the final feature matrix by removing unnecessary columns.

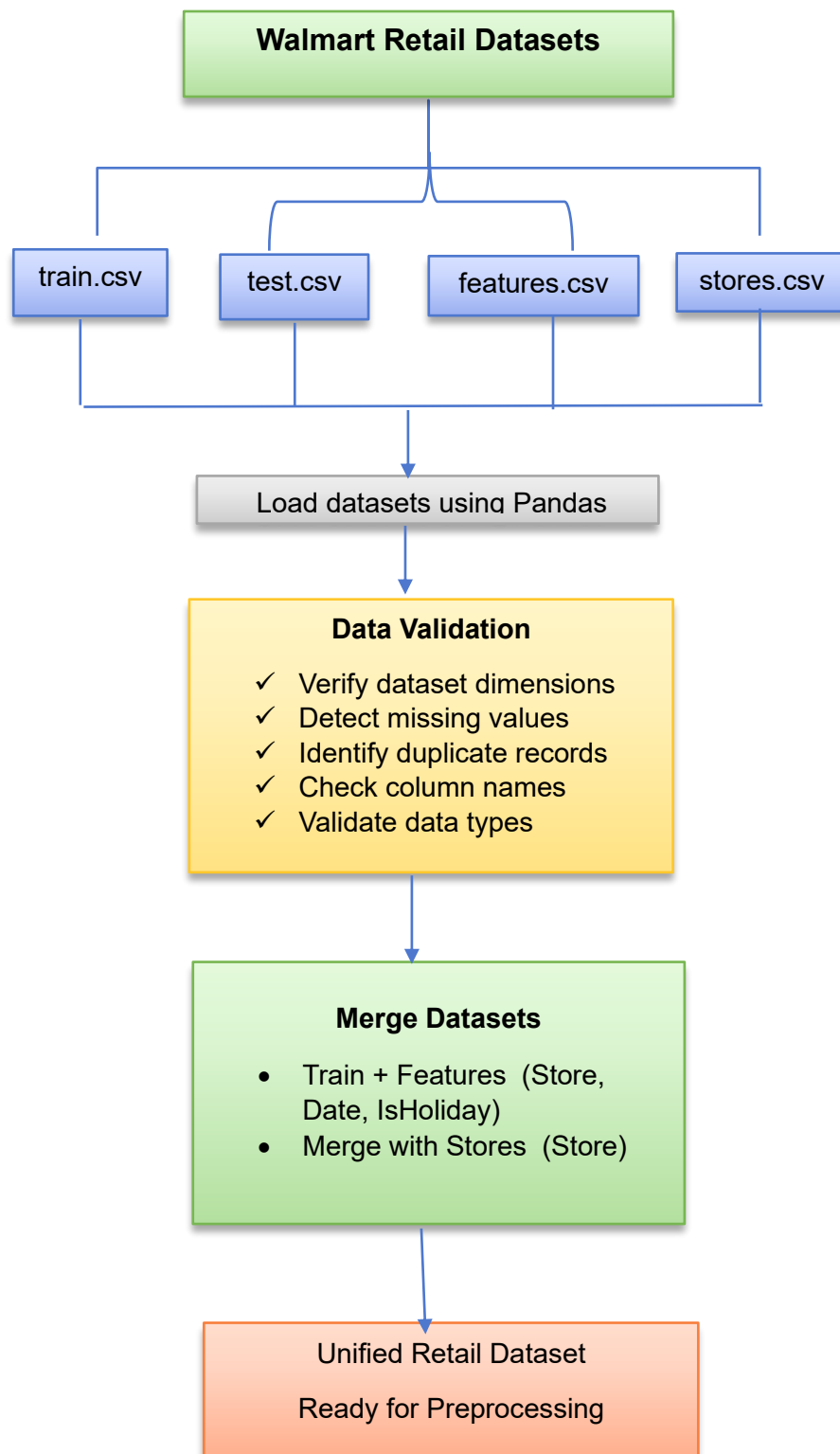
### 4. Model Training & Validation

Trained **Linear Regression**, **Random Forest**, and **XGBoost** models. Optimized model hyperparameters using **Optuna**. Evaluated model performance using **MAE**, **RMSE**, **MAPE**, and **R<sup>2</sup> Score**. Selected the best-performing model based on validation results.

### 5. Prediction & Post-Processing

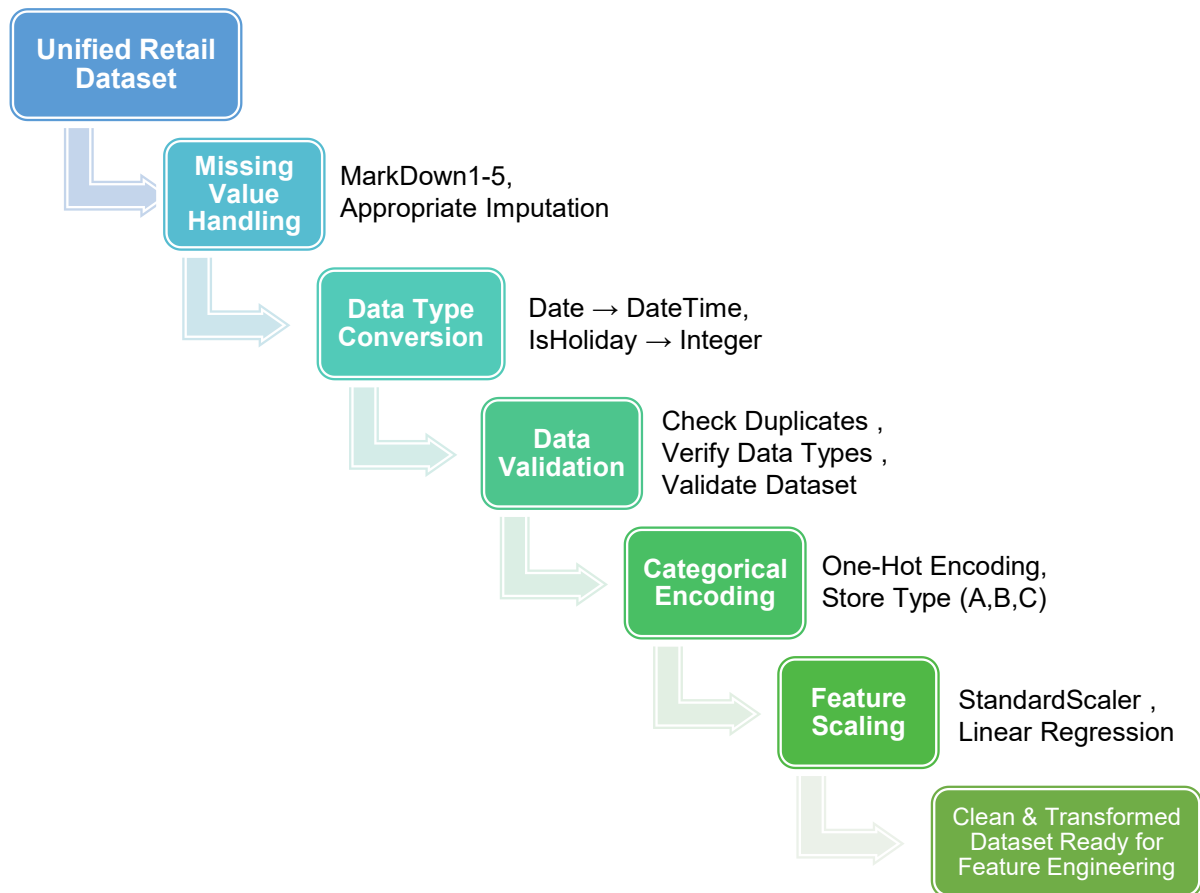
Generated demand predictions using the selected model. Assessed prediction accuracy on the validation dataset. Applied **SHAP** to interpret global feature importance. Used **LIME** to explain individual model predictions. Saved the final model and generated prediction results.

## 1 . DATA INGESTION PIPELINE



The data pipeline process is kicked off by reading in the four Walmart datasets into a pandas DataFrame. Before any more processing or manipulation, each dataset is analyzed to make sure it is ready for processing and feature engineering (i.e. Checking shape of the data, column names, data types, checking for missing values, and duplicate records). Then, all four datasets are merged together based on Store, Date, and IsHoliday (all 4 tables shared these 3 columns) to make up one big data frame. This combined data frame contains store information, historical sales data, and other external business information which will form the basis for all further processing and analysis.

## 2. DATA CLEANING & TRANSFORMATION PIPELINE

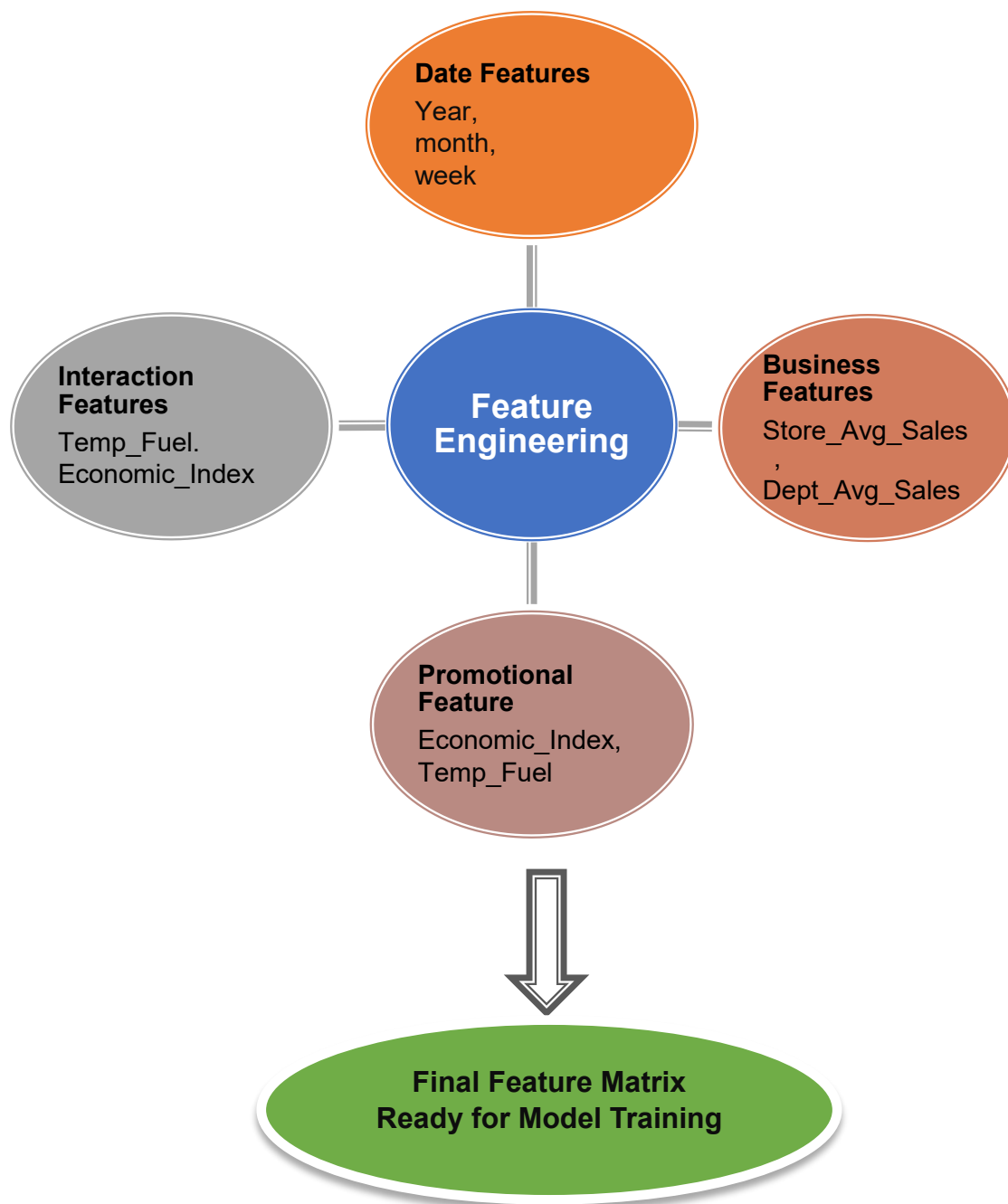


Cleaned and Transformed Unified Retail Dataset to have quality data for Machine Learning. Missing values present in Markdown Features are imputed by a suitable value to retain important information of promotions. Date feature is converted into datetime format for time based feature extraction. The IsHoliday column is converted into numerical format.

Duplicated records were verified in the dataset along with checking data types.

Store Type(Categorical feature) is encoded using One-Hot Encoding and the numerical features are scaled using StandardScaler since there are algorithms requiring features to be scaled. The above mentioned clean dataset is used for the next step i.e., Feature Engineering.

### 3. Feature Engineering Pipeline

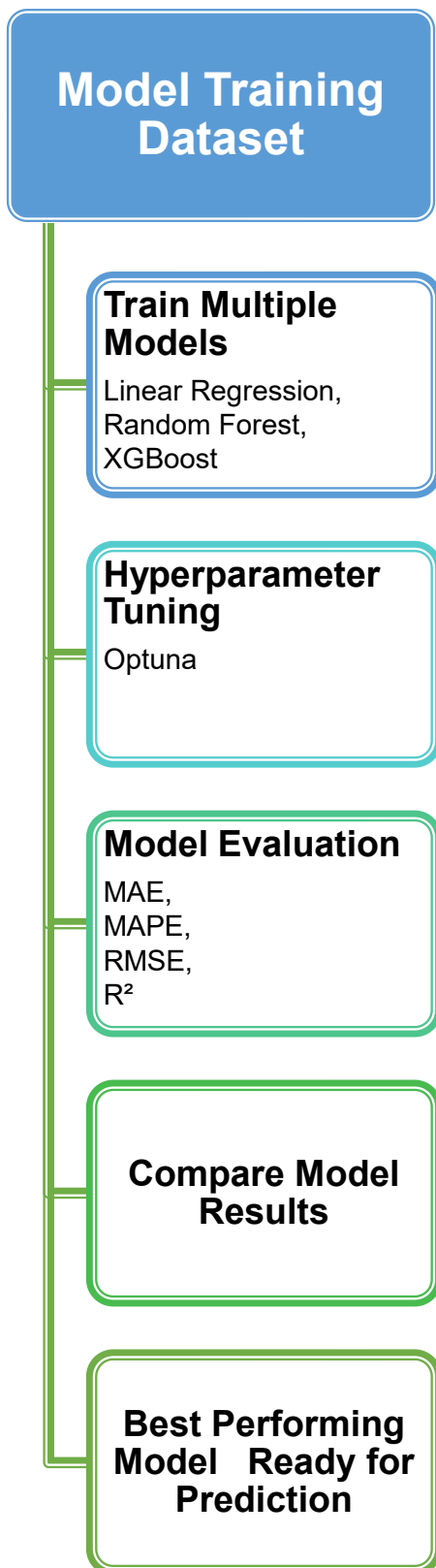


**Feature Engineering** We implemented feature engineering on the cleaned data to develop new features that would better enable the model to capture trends in time, the business cycle and sales behavior from past periods. We extracted time-based features that included Year, Month and Week to capture seasonality and time based effects. We developed business level features such as StoreAvgSales and DeptAvgSales, which represent historical performance at the store and department level, respectively.

We generated TotalMarkdown by aggregating all the markdown variables to capture the promotional effect and TempFuel and Economic\_Index were created to capture effects of the environment and economy and their interactions.

Finally, we dropped few columns to develop the final feature set.

## 4. Model Training & Validation

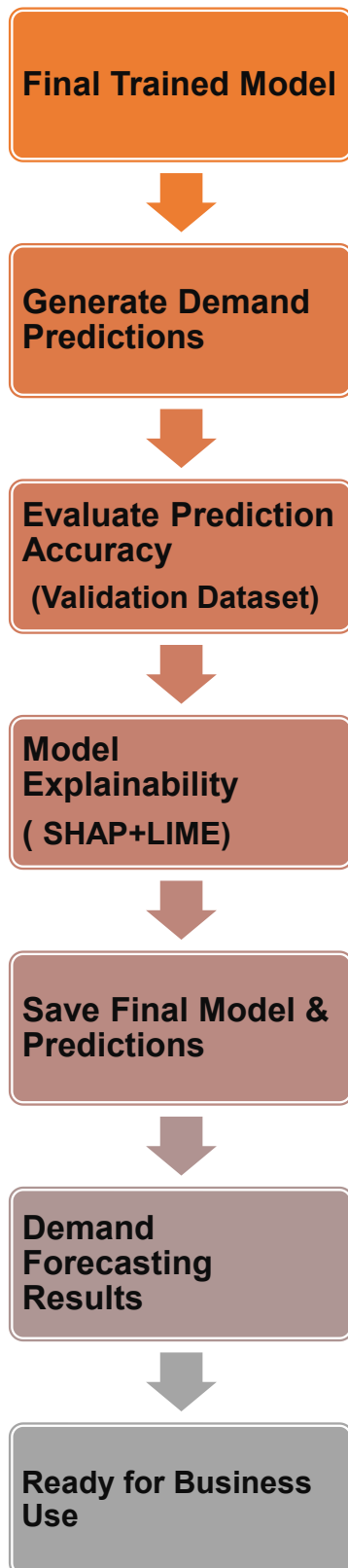


In this section of the study, the main goal was to obtain reliable demand forecast models from the developed dataset. As explained above, machine learning techniques have a strong capability in modeling trends present in historical sales data. For this purpose, four different machine learning algorithms have been used to generate forecasts: Linear Regression, Random Forest, and XGBoost.

All the models have been optimized with Optuna.

For evaluation of each model, mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and R score are the primary metrics to calculate for prediction error. The best performing model, from among the above mentioned algorithms

## 5: Prediction & Post-Processing



The prediction and post-processing phase creates the demand forecast based on the selected machine learning model and confirms the forecast accuracy on out of sample data. Explanation methods (SHAP, LIME) of global feature importance and individual predictions help explain the model results. Lastly, the trained model and forecasts are stored to be used in future forecasts, analysis and business decisions.